

VXSlate: Exploring Combination of Head Movements and Mobile Touch for Large Virtual Display Interaction

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Virtual Reality (VR) headsets can open opportunities for users to accomplish complex tasks on large virtual displays using compact and portable devices. However, interacting with such large virtual displays using existing interaction techniques might cause fatigue, especially for precise manipulation tasks, due to the lack of physical surfaces. To deal with this issue, we explored the design of *VXSlate*, an interaction technique that uses a large virtual display as an expansion of a tablet. We combined a user's head movements as tracked by the VR headset, and touch interaction on the tablet. Using *VXSlate*, a user head movements positions a virtual representation of the tablet together with the user's hand, on the large virtual display. This allows the user to perform fine-tuned multi-touch content manipulations. In a user study with seventeen participants, we investigated the effects of *VXSlate* on users in problem-solving tasks involving content manipulations at different levels of difficulty, such as translation, rotation, scaling, and sketching. As a baseline for comparison, off-the-shelf touch-controller interactions were used. Overall, *VXSlate* allowed participants to complete the task with completion times and accuracy that are comparable to touch-controller interactions. After an interval of use, *VXSlate* significantly reduced users' time to perform scaling tasks in content manipulations, as well as reducing perceived effort. We reflected on the advantages and disadvantages of *VXSlate* in content manipulation on large virtual displays and explored further applications within the *VXSlate* design space.

Additional Key Words and Phrases: virtual large displays, VR, mobile device, touch interaction, head movements

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1 INTRODUCTION

Large displays, either in the form of a single large screen or multiple tiled screens, have been widely employed to support complex problem solving tasks. Typical domains where practitioners need to deal with complex information manipulations and decision-making are emergency response [23], traffic management [24, 66], visual data exploration [2, 41], medical imaging diagnosis [2, 41], and product design [19, 20, 70]. Users can perform interaction techniques on large displays, such as selecting nearby objects, moving, scaling and rotating contents using bodily gestures [37, 85],

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remote pointing [58, 61], or using handheld devices [12, 14, 21] for sketching with direct touch interaction [33, 62]. However, large physical displays also constrain users to stationary and bulky setups, which are quite impractical for home users or for travelers who need to perform such tasks while being out of their office or workplace.

Recent progress in Virtual Reality (VR) provides users with the possibility to view and interact with virtual high-resolution contents using a compact hardware setup. Inherently, this opens new opportunities to develop virtual environments consisting of large virtual displays. These displays allow performance of complex tasks without the need for large physical displays and with potential benefits for medical image diagnosis [26, 92], emergency response [8], and collaborative design [27]. Here, we define "large virtual displays" as displays being projected onto the wall of a VR room viewed through a VR headset. These displays do not necessarily have a physical representation in the real world and thus do not need a physical space. Using a VR headset, current VR technologies can easily allow users to view virtual displays that are even larger than physical ones. This is combined with a much lower cost for hardware infrastructure. Using VR, individuals in out-of-office contexts can still easily work on large virtual displays without being physically constrained to certain locations. In addition, using large virtual displays in VR can also overcome ambient illumination and screen-reflection issues often associated with physical displays [77, 92]. However, a major weakness of large virtual displays is the lack of a physical surface to allow for intuitive interaction and passive haptic feedback. This might lead to user fatigue since users have to interact using mid-air gestures without or even with controllers. Although there have been efforts to deal with fatigue by affixing virtual displays to nearby physical surfaces such as walls or tables to provide passive haptic feedback [22, 95], these room-scale approaches might lead to spatial conflicts between users and other people collocated in the same space [88, 97], or might even be socially unacceptable in public places [71].

Mobile devices like tablets provide multi-touch interactions on a medium-sized surface and are now being used for precise distal content manipulations on a small part of a large physical display [57]. Touch tablets can be also used to support precise interactions, such as sketching in Mixed Reality (MR) [96] or rotating and scaling 3D objects in VR [81]. Thanks to passive haptic feedback from tablet screens, such devices cause less user fatigue [89]. Also, interactions on their relatively small screens provide more comfort than mid-air interactions [89]. However, this can result in imprecise interactions on large physical displays when simply up-scaling and mapping the interactions on the tablet's screen. To tackle this problem, head movements are tracked and coupled with the physical tablets allowing users to manipulate contents on portions of large physical displays [57, 82] where the user is looking. With current VR systems, facing direction angles (here: right-left and up-down) can easily be acquired by applying ray casting from the user's first-person virtual camera (also referred to as head gaze [5, 11]). This gives access to a combined use of facing direction and multi-touch interaction to potentially give users access to large virtual displays in VR using off-the-shelf hardware. However, it is yet unclear how this interaction technique affects users in complex content manipulations.

In this paper, we explore *VXSlate* (Virtually eXtendable Slate), a design that considers a large virtual display as an extended surface of a tablet. The design combines head movements, tracked by an off-the-shelf VR system with input on a touch tablet. This allows a user to perform various manipulations on a large virtual display ranging from translation, rotation, and scaling of objects to sketching. *VXSlate*'s design was built based on previous work on interacting with large physical displays combining head movements and touch. However, *VXSlate* is different because it combines a user's head movement with touches on a tablet to interact with a large virtual display which is seen through the VR headset. This combination offers a compact setup that allows users to interact with large virtual displays more flexibly than with systems relying on large physical displays. We then refined it through an iterative process taking into account users' interactions in VR [81, 99]. *VXSlate* differs from other systems by using the user's head movements exclusively for positioning the virtual representation of the tablet on an area of interest in the large virtual display. This makes use

of spatial information of touches on the tablet for content selection and manipulation, providing users the experience of touching target contents.

In a user study, we examined the effects of *VXSlate* on users' performance and experience in comparison with off-the-shelf VR touch controllers. Seventeen participants each had to perform a problem-solving task consisting of matching jigsaw-puzzle pieces and sketching, thus involving content manipulations at different levels of difficulty. The study's results show that by using *VXSlate* participants complete the task with completion times and accuracy comparable to using touch controllers. Moreover, *VXSlate* can significantly reduce the time to perform certain fine-tuned content manipulations, such as scaling. After an interval of use, *VXSlate* reduces a user's perceived effort for completing the task. Finally, we discuss potentials of exploring the *VXSlate* design space and potentials for mixed-reality (MR) uses.

2 RELATED WORK

VXSlate was built on the basis of work ranging from interactions with large physical displays to interacting with virtual contents in VR using physical surfaces or gaze. In this section, we will explain these interactions in more detail. Finally, we distill our design rationale for *VXSlate* from related research.

2.1 Large physical display interaction

To reduce physical navigation in interacting with large physical displays, there are several approaches that allow users to distantly manipulate contents. One of the earliest approaches was to support distant pointing and selection tasks on large physical displays by applying ray-casting on laser pointers [10, 56], on users' pointing gestures or on pointing sticks [42, 86]. For settings where a user's body casts a shadow, this could be used to facilitate manipulation over large distances [75, 76]. Brasier et al. [17] explored an approach that redirects a user's below-waist and near-body mid-air hand gestures to distantly interact with a large virtual display in pointing and steering tasks. Hardware development for smartphones opened opportunities for users to select and manipulate contents on large physical displays using handheld devices, utilizing their built-in camera [13], touch screen [54, 72], keyboards [34], and inertial motion sensors [15, 64]. Research has mapped touch interaction on tablet-size pads onto large virtual displays for manipulations. Both Malik et al. [51] and Nancel et al. [57] explored techniques that allow mapping a user's touch interaction for absolute and relative navigation onto large virtual displays. The system from Malik et al. [51] in particular allows multi-touch interaction for indirectly grouping, zooming, and scaling contents on the large physical display. This system also captures and shows images of the user's hands interacting with the pad on the large display to raise the user's awareness of what his/her hands are doing. The orientation of the handheld devices (e.g. smartphones, tablets) can also be used to determine the interaction area on the large display. Multi-touch gestures on these devices are used for content manipulation, for instance selecting, dragging, dropping, scaling and rotating [50, 57].

Combinations of a facing direction and touch interactions on mobile devices have also been used for remotely manipulating contents on large physical displays. The results of several studies [57, 78, 79, 82, 84] typically make use of facing direction or eye gaze as a preliminary selector, roughly determining a target on the large physical display. Touch interaction was considered as indirect and was mapped to objects being selected by facing direction or eye gaze. Also, the user did not need to watch touch positions of his/her fingers on the tablet.

In highly precise tasks, like tracing or sketching, the use of the facing direction or eye gaze can be problematic. Unintentional swaying head movements can also easily lead to errors for such highly precise tasks. Similarly, using eye gaze as input to steer a drawing process can be tedious due to effects of saccades. Eye gaze is highly coupled with the movement of the fingertip or the pen tip that is creating the drawing [30]. Thus, a user will need to align his/her finger

directly along the path of the drawing, triggering touch interactions on the large virtual display. Using this technique, where eye gaze acts as an object selector and touch interactions are for object manipulations, the user needs to switch between an indirect touch interactions mode in selecting, rotating, or scaling objects to a direct mode in tracing or sketching. Nancel et al. [57] argued that such switching should be minimized to reduce cognitive workload on the user.

2.2 Physical surface interaction in VR/MR

To reduce fatigue caused by mid-air interactions in VR, physical surfaces have been used to provide the user with passive haptics in certain tasks. More specifically, tablets and sketching pads were utilized to reduce user fatigue in docking [49] and sketching tasks [96] in VR or mixed reality (MR). Researching hand-held windows for 2D interaction, Lindeman et al. [49] showed how passive haptic feedback can support precise movements in virtual environments.

Arora et al. [4] showed that sketching in VR, using a physical surface, outperformed mid-air interactions. They suggested the use of a mobile surface controlled by a robot arm to assist users in sketching in VR. However, this suggested setup might be bulky and unfeasible for traveling individuals. Drey et al. [25] explored VR sketching, where a user's input on a tablet screen is mapped onto a larger virtual display in VR. When the user needs to perform more fine-tuned sketching, he/she performs that directly on the virtual content using mid-air pen gestures. This might lead to fatigue and certain social unacceptability in environments where physical navigation in the environment is not appropriate. MRTouch [95] turned any physical surface around the user into touch displays in MR. However, this approach might not be socially acceptable in public places. Besides sketching, physical surfaces like tables were also effectively employed to assist radiologists in interacting with medical images in VR [77, 93]. Also addressing the medical domain, Sundén et al. [80] explored interactive multi-touch tables for navigating scientific data sets.

Surale et al. [81] used a tablet's orientation to select 3D objects in VR and utilized touch interactions on the tablet to scale and rotate 3D content. The user can also draw text on the tablet and then scale, rotate, and position the text in the 3D space. BISHARE [99] allowed users to select, position, and rotate 3D objects in MR using a phone's orientation. In addition, users could perform fine-tuned manipulations, such as scaling or changing colors, on the object using touch interactions on the phone. However, these approaches might not be suitable for highly precise tasks, for example when tracing or sketching directly onto a large virtual space. Also when switching between selection and manipulation tasks, the user needs to change the way he/she holds the tablet, which can be disruptive to his/her workflow.

Further studies on physical surface interaction with VR/MR include Feuchtnner's and Müller's [28] mid-air hand gesture technique, showing ways to reduce physical stress. Dollhouse [40] combined VR and tabletops for team members with complementary roles, where the VR users had frontal touch input on their headset. FaceDisplay even proposed a VR headset with both touch input and mobile displays for surrounding non-VR users [31]. Slicing-Volume [55] (a tablet-based 3D selection technique for dense VR environments) and SymbiosisSketch [3] (combining mid-air 3D drawing with stroking on a 2D tablet), are also of relevance to our research.

2.3 Eye gaze and head movement interaction in VR/MR

Since the early days of VR/MR technologies, facing direction has been intensively researched to aid users' interaction with virtual contents. Ohshima et al. [60] instrumented a user's eye gaze to strategically render in-focus contents in VR, mimicking humans' central-peripheral vision. Based on theories of human attention [29, 73, 83], gaze has been widely used for target selection in VR/MR [46, 69, 98]. In low-cost mobile VR, target-selection techniques are often based on facing direction. Qian et al. [67] showed that even target selection in VR using facing direction is more accurate than

techniques based on eye gaze. Pfeuffer et al. [63] extended this interaction technique by allowing users to use mid-air hand gestures to select and manipulate virtual objects.

2.4 Design rationale for VXSlate

From the above literature review, we derive the following design rationale, presented in three aspects: *First*, since eye gaze is an error-prone input technique for tasks where setting the area of interest is combined with precise input, we propose to use facing direction instead of eye gaze. *Second*, to our knowledge, there is so far no work combining facing direction and touch where the large virtual display is considered as an extended space of a handheld device. Previous studies on facing direction-touch interaction for remotely manipulating contents on large virtual displays mainly focused on docking tasks (including selecting, positioning, scaling and rotating), and less on highly precise ones like tracing or sketching. In *VXSlate* touch input at the physical tablet is visually mapped onto the virtual tablet, allowing the user to refine selection, performing object manipulations like scaling, rotating or moving a short distance, or they can be combined with head movements to quickly move a selected object across a long distance and the large virtual screen. *VXSlate* complements this research direction by introducing a technique in which head movement determines absolute navigation and touch interaction aims to provide a less fatiguing way for relative manipulations on the in-focus area. *Third*, *VXSlate* aims to remove the mode switching issue of touch interactions.

VXSlate extends previous studies by providing an approach to interact with large virtual displays in VR (and also applicable to MR) through touch interaction on a small-sized surface like a tablet, combined with head movements designed to reduce user fatigue while maintaining a compact setup. For various tasks with varying levels of precision, our work includes a comparative experiment between a) *VXSlate* and b) mid-air interactions using touch controllers.

3 DESIGN OF VXSLATE

VXSlate was designed to support a wide range of content manipulation tasks, from selection, moving, scaling, and rotating objects to fine-tuned interaction, such as sketching or tracing. These are tasks that users can perform on a large physical display. Thus, we argue that to foster the use of large virtual displays in VR, it is necessary to support users to perform all manipulations. In addition, *VXSlate* supports passive haptic feedback. *VXSlate* is designed as a portable setup where the user is spatially unconstrained. To achieve this design, *VXSlate* uses the touch interaction on a tablet. A tablet is portable, yet offers rich multi-touch interactions; the user can perform a wide range of object manipulations, such as selecting, scaling, and rotating. Touch interactions on the tablet screen also allow the user to perform highly precise tasks, like sketching or positioning. Since the tablet screen is much smaller than the large virtual display, only a simple absolute mapping of touches on the tablet to the large virtual display would be imprecise. Inspired by previous work on large physical display interaction [32, 38], we chose to combine touch input on the tablet with head movements.

In our approach, head movements are used to set a particular area of interest on the large virtual display. Touch interactions on the tablet are mapped onto that area to manipulate contents. More specifically, interactions using *VXSlate* include two phases: head movements determine the area of interest on the large virtual display, while touches on the tablet are mapped onto the area of interest and directly control contents on the large virtual display. Figure 1 illustrates the concept of *VXSlate*.

3.1 Head movement defining the area of interest

The user's area of interest on the large virtual display, denoted as the *virtual tablet*, is represented as a rectangular area congruent to the tablet. Unintentional head movements might cause frustration due to an unstable *virtual tablet*'s

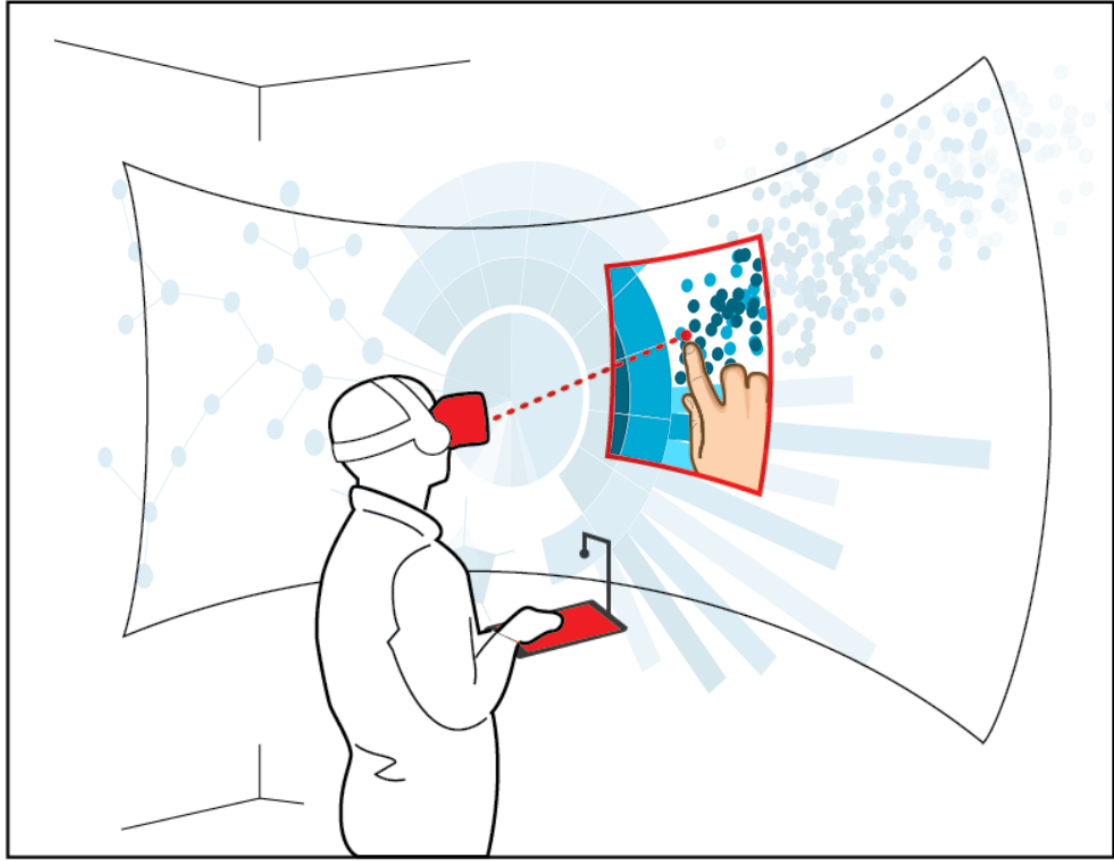


Fig. 1. VXSlate concept: The user wears an HMD, seeing a large virtual display in a virtual room. The user's hand interactions on and above the tablet are captured by a camera mounted above. They are then visually mirrored on the *virtual tablet* (the red rectangle) shown on the large virtual display (in the prototype implementation, this is shown as a yellow rectangle for better contrast in the virtual environment). The *virtual tablet* is placed by user head movements.

position, especially in highly precise manipulation tasks, like accurate tracing and positioning. Thus, we needed to reduce movements of the *virtual tablet* caused by unintentional swaying. After several pilot tests, we chose an approach detecting head movements that are likely to be intentional ones. In this approach, we consider angular velocities of head movements in the latest 10 frames. If the average magnitude of angular velocity is greater than around 75 degrees per second and the average change in angular velocity between two consecutive frames is greater than around 0.5 degrees per second (at a 60fps framerate in our prototype), head movements are considered to be intentional. The threshold values were derived through our pilot testing. In our preliminary testing, we found this approach was sufficiently robust to stabilize the *virtual tablet* while the user performed precise manipulation. These values were then used for the user study. When intentional head movements are detected, the *virtual tablet* transitions to a new position.

While working with large virtual displays, the user might have various preferences between the precision and the responsiveness of the system. For instance, when moving an object using touch interactions, the user typically desires

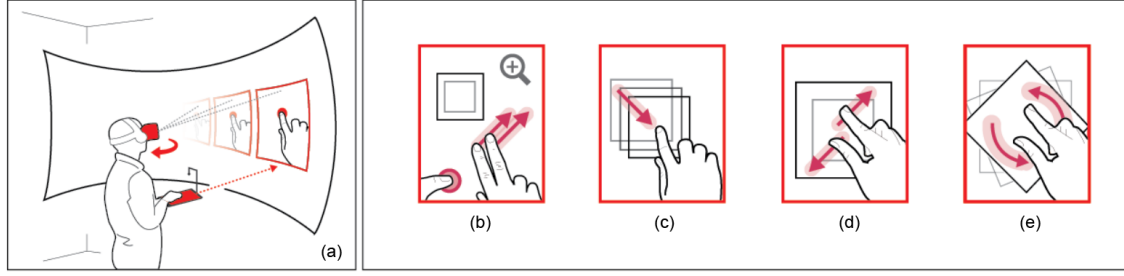


Fig. 2. VXSlate touch interaction. (a) Quickly moving an object as well as the *virtual tablet* across the large display by holding a finger touching the object and orienting facing direction towards the target area. (b) Resizing *virtual tablet* by moving two fingers of one hand while holding the tablet with the other thumb on the screen. (c) Selecting and moving/positioning an object using a finger touching it. (d) Scaling an object using a two-finger pinch gesture. (e) Rotating an object by a two-finger gesture.

to drag it quickly to the target area. whereas when tracing a line, the user aims for precise touches. To support this, VXSlate allows the user to resize the *virtual tablet* proportionally to adapt to their task: larger for faster interactions on a large area, and smaller for tasks focusing on high precision. The user can resize the *virtual tablet* using the following bi-manual touch gesture: holding the thumb (of the hand holding the tablet) on the screen and moving two fingers of the other hand closer together or further from the thumb to scale the *virtual tablet* up or down respectively (see Figure 2b). This gesture mimics the common two-finger pinching gesture for object-scaling on mobile devices; thus, it should be easy to use.

3.2 Touch interaction for direct content manipulation

Different from previous work, VXSlate takes into account touch positions on the tablet to make use of direct-touch interactions for manipulations of distal contents on large virtual displays. Users' touches on the tablet screen are congruently mapped onto the *virtual tablet*. The touches will directly affect the content on the large virtual display. The user thus can easily apply his/her interactions on commodity mobile devices to work with the large virtual display VXSlate. More specifically, to select an object, the user just needs a single touch on it via the tablet (see Figure 2c). Likewise, to scale or rotate an object, the user can perform two-finger pinching or two-finger rotating gestures on the tablet screen (see Figure 2d, 2e). Similarly to sketch or trace, the user can touch on the position on the large display to create the drawing via the tablet and directly perform the task there. We argue that this approach should resolve the issue of switching between direct and indirect touch in gaze-touch interaction raised by Nancel et al. [57].

To move an object, the user can directly select and drag it within the *virtual tablet* via touch interactions on the tablet (see Figure 2b). VXSlate provides hybrid interaction combining touch and head movements allowing users to quickly move an object across the large display. To do that, the user selects an object with a one-finger touch on the tablet. The selected object then moves together with the *virtual tablet* to a new area, where the user can refine the positioning. The selected object always maintains its relative position to the *virtual tablet* and to the touch finger (see Figure 2a). This hybrid approach allows for quick head movements while maintaining direct touch.

As touch positions are important in the direct touch interaction of VXSlate, it is necessary to visualize them sufficiently on the *virtual tablet*. In our prototype, a touch on the tablet screen is represented as a red circle at its corresponding position on the *virtual tablet* (see Figure 2). While wearing a VR headset, the user cannot see his/her hands on the tablet. Therefore, it is necessary to represent user hands within VR so the user can locate them on the virtual tablet display

before touching it. While the hands interact with the tablet, the user’s visual attention is focused on the large virtual display. The hand movements are highly dynamic as they switch between various gestures. Their representation on the *virtual tablet* mirrors these gestures.

In a pilot study during the design process, we found that using short-range hand-capture devices like Leap Motion mounted on top of the tablet could not provide reliable tracking. Furthermore, glove-based solutions for finger-tracking are not easily available. We thus employed a simple approach similar to the work of Malik et al. [51] and Le et al. [48] for capturing the user’s hand interactions on and above the tablet screen. A conventional camera is mounted above the tablet to capture its screen. The captured images are processed to remove the background, retaining hands only. This will be visually mirrored on the *virtual tablet* (see Figure 1). It should be noted that this setup serves the sole purpose of capturing the images of the user’s hand interactions on and above the tablet’s screen and presenting them on the *virtual tablet*. The user’s touch interactions on the tablet are still captured by the touch capability of the tablet’s screen, using supporting APIs. Since such images do not convey depth information, when a finger touches the tablet screen, its location is highlighted by a red circle on the *virtual tablet*.

In pilot testing, we found that while detection of intentional head movements worked well in most cases, there were still cases of unintentional movements of *virtual tablet*. We found that while these movements were almost unnoticeable to the users during positioning tasks, they led to reported frustration during scaling, rotating, and tracing. When a user starts manipulating objects using touch interactions, their focus will be on the touches rather than on their head movements. To avoid faulty interactions, potentially caused by unintentional head movements, the *virtual tablet*’s position will be locked during touch interactions. The *virtual tablet*’s position will be unlocked to follow the user’s head movements when there are no touches on the tablet or when the existing touches stand still.

3.3 Implementation

VXSlate was implemented using Unity on a Windows computer combined with an Android application running on a tablet. During the prototyping process and evaluation, we used an Asus Nexus 9 tablet equipped with a 8.9-inch screen, 1536×2048 resolution and running Android 8. The tablet was used in the portrait mode as this caused less fatigue than the landscape mode when the user needed to hold it with one hand. Users’ touch interactions are detected by the tablet screen and images of hand gestures are captured by a camera connected to the tablet. The tablet and the Windows computer communicate with each other via a local wireless network. The main screen of the Android application is blue (RGB = 0,0,255) to make it easily distinguishable from skin colors for background segmentation. When a user’s fingers touch the tablet’s screen, each touch point will be highlighted by a cyan color (RGB = 0,255,255) to inform the user that his/her touches on the screen can be captured by the application. The Android application captures information from the user’s touches on the tablet screen via touch events supported by Android APIs and sends it to the Unity application in the form of byte-array data. We 3D-printed a plastic arm to mount a Logitech C615 webcam 30 cm above the tablet screen. As this is a proof-of-concept, we connected the webcam directly to the Windows computer in order to reduce the latency of image streaming to the Unity application. Images captured by the webcam are processed using a simple color-based background segmentation to keep only the skin-color elements ($Y \geq 10, 132 \leq Cb \leq 172, 77 \leq Cr \leq 127$) [68]. They are then sent as a MJPEG stream to the Unity application. Even though we used an external camera in this prototype, the development in wireless transmission should allow for more portable and light-weight solutions. An early advance in this direction can be found in MirrorTablet [48], which added a built-in front-facing camera hinged onto the tablet. Following MirrorTablet, images of the user’s hand are visually mirrored onto the *virtual tablet* with an opacity of 70%. This is sufficient for the user to see both his/her hands and the large display’s contents [35]. The

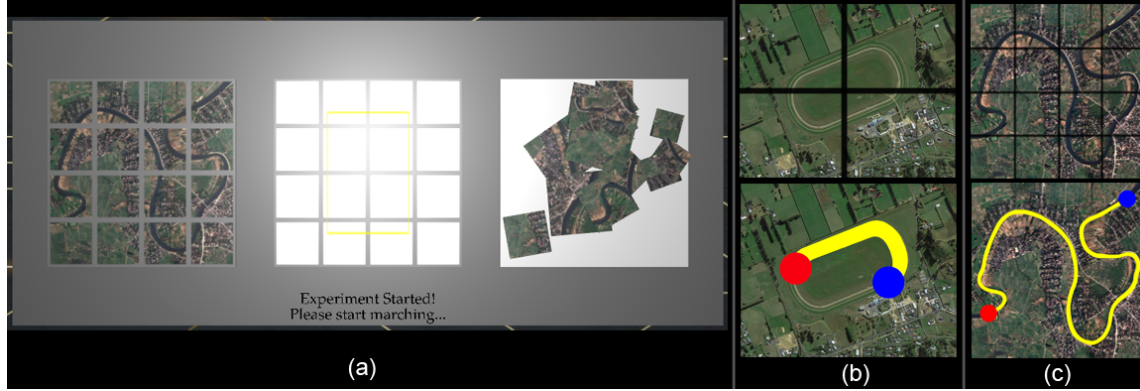


Fig. 3. (a) Large virtual display setup in the study: Left - the template puzzle, Middle - Puzzle solving area where participants need to correctly place puzzle pieces to assemble the template puzzle, Right - pool of puzzle pieces. (b) 2x2 puzzle used in training: Top - puzzle template with grid lines for *Matching stage*. Bottom - puzzle template with predefined path and without grid lines for *Sketching stage*: Red circle indicates starting point and blue circle indicates end point; positions of the two circles are swapped for reverse direction. (c) 4x4 puzzle used in the experiment: Top - puzzle template with grid lines for *Matching stage*. Bottom - puzzle template with predefined path and without grid lines for *Sketching stage*

highlighters of the user's touches on corresponding positions on *virtual tablet* are red circles with 50% opacity, allowing them to see the contents behind.

4 EVALUATION

We conducted a user study to examine the effects of VXSlate on users' performance and perception in performing manipulation tasks with different levels of difficulty.

4.1 Experiment design and task

We carried out a between-subject study comparing the VXSlate technique to a classical interaction technique based on the use of *touch controllers*. This study design was chosen to avoid task learning effects, which were observed in our pilot study. We chose to use touch-controller interaction as a baseline because this is still the most common technique in commercial VR systems. Interactions using touch controllers are also more reliable than marker-less, bare-hand interactions.

The experimental task in the study was a game that we designed based on problem-solving activities commonly deployed on large physical displays. Through surveying the literature [2, 19, 23, 24, 41, 66], we ranked the content manipulation tasks according to level of difficulty: selecting, docking (including moving, scaling, rotating, and placing), and steering [9]. Selecting is when a user targets an object before performing further manipulations; it is considered to be the simplest canonical task [9]. Docking tasks are when the user moves the selected object, resizes it, adjusts its orientation, and positions it [90] (e.g. designers moving sticky notes to their corresponding group, rotating them, or making them larger for readability). Docking tasks are considered to be more challenging than selecting tasks [9]. Steering tasks are those that require input from users following certain trajectories, such as writing, drawing, and navigation [1]. In problem-solving tasks, steering can be found in various scenarios, for example pathologists sketching or annotating medical images, or traffic planners drawing lines on a map.

While previous studies required participants to perform content manipulation tasks separately, in our study all of these tasks were part of the problem-solving game that participants were asked to play. The intention was that in our study design, the primary focus of participants should be to overcome the challenges of the game, with only secondary focus paid to manipulation tasks. Our design helped in evaluating the effects of the interaction techniques (*VXSlate* versus touch-controllers).

We designed a problem-solving game consisting of two stages for on a large virtual display. In the first stage, the user needs to solve a jigsaw puzzle by placing puzzle pieces to assemble a predefined map image. This stage is referred to as the *Matching stage*. In this stage, the large virtual display is vertically divided into three areas (see Figure 3a). The *left area* shows the *template of the puzzle* that participants need to assemble. The puzzle template is divided into a grid of rectangular pieces to let participants easily recognize the pieces needed for the matching task. The *center area* is the *matching area* where participants need to correctly place, scale, and rotate puzzle pieces to their correct position, size, and orientation. To help participants easily determine the position of puzzle pieces, this area is also divided into the same grid of rectangular cells as the puzzle template. Each puzzle piece only fits one cell in the grid. A puzzle piece is considered to fit if: the position deviation of its center is less than 5% of the cell's diagonal, its rotation deviation is less than 5 degrees, and its size deviations are less than 5% of the cell's size. Through pilot testing, we found that these values made the matching adequately challenging for participants, but not so difficult as to cause frustration. When a puzzle piece fits its corresponding cell, its frame will be highlighted in yellow. The puzzle is solved when all the pieces are correctly matched. The *pool of puzzle pieces*, from which participants pick a piece for matching, is the area on the right. When the game starts, puzzle pieces are stored in the pool at random positions, with scales randomized from 50% to 150% of the corresponding cells, and at random rotation angles. Participants select a piece, drag it to the matching area, and adjust its position, size, and orientation. Once a piece is selected, it is highlighted by a red frame. In this *Matching stage*, we evaluated the performance of *VXSlate* in selecting and docking tasks (moving, scaling, rotating, and placing).

Once a puzzle has been solved, the game proceeds to its second stage, called the *Sketching stage*. Here, participants trace a predefined path on the puzzle image they have just assembled in the *Matching stage*. The purpose of this task is to evaluate *VXSlate* for steering tasks, which are more difficult than docking tasks. The steering task imitates annotation in various problem-solving scenarios. We intentionally chose map images as puzzle templates for the game, so that the sketching task could be designed as tracing along streets or rivers. To finish the *Sketching stage*, participants begin tracing at a starting point (indicated as a red circle $(R, G, B) = (255, 0, 0)$) and then follow as precisely as possible a predefined path depicted by the color yellow $((R, G, B) = (255, 255, 0))$ until they reach an end point (indicated as a blue circle $((R, G, B) = (0, 0, 255))$). Once the end point has been reached, the starting point and end point are automatically swapped and participants retrace the path in the reverse direction. Requiring participants to perform two-way tracing helped us reduce the biases caused by cases where some people are better at navigation in a certain direction. Since we focused on investigating the effects of *VXSlate* when participants were performing a high precision task, we instructed them that it was more important to trace precisely rather than quickly. Participants were also informed that it was not necessary to trace the entire path in one continuous line; they could halt their tracing line and then re-start. For the steering task, the assembled puzzle was replaced by a corresponding template without grid lines, making tracing easier.

4.2 Apparatus

The experiment was carried out in a closed 5×5m physical room. The participants were seated on a 50cm height chair located 2m from the computer running the VR game. The virtual environment was created and rendered by Unity3D on

a Lenovo ThinkStation P310 computer. The computer was equipped with a 6th generation Intel Core i7-6700 (4 cores, 3.4GHz, 8MB, 65W, DDR4-2133) processor, 32GB of RAM, and an NVIDIA Quadro K420 graphics card. An Oculus Rift S head-mounted display (HMD) was used for both of the techniques. The HMD provides an 1080×1200 pixels OLED display per eye with a 110 degrees field of view, and a refresh rate of 90 Hz. The positions of participants and touch controllers were tracked by the built-in cameras of the headset. For *VXSlate*, we used the same tablet and camera as in the prototyping process.

The virtual environment was a 10×10×4m room. The game was shown on a flat 7×3m large virtual display placed at one side of the room. The participant was positioned at the center of the room, 5m away from a large virtual display. By default, the *virtual tablet* was displayed at the dimensions of 1×1.3m. We used one puzzle template for all participants in the study (see Figure 3c). For measuring tracing accuracy, we chose puzzle templates that did not contain pixels outside the predefined path having one of the three colors: red ((R, G, B) = (255, 0, 0)), blue ((R, G, B) = (0, 0, 255)), and yellow ((R, G, B) = (255, 255, 0)). In the user study, we used a 4×4 puzzle (see Figure 3c). The three areas (Puzzle Template Area, Matching Area, Pool of Puzzle Pieces Area) on the large virtual display had equal sizes of 1.848×1.848m. Both the Puzzle Template Area and Matching Area were divided into a 4×4 grid. Each grid cell measured 0.462×0.462m.

Touch controllers are motion controllers accompanying the Oculus Rift S VR headset. There are two controllers, designed for left- and right-hand usage. They provide users with six degrees of freedom interactions. Each controller has three buttons, one thumbstick with one grip, and one front-facing trigger key. The controllers provide haptic feedback, and are powered by AA batteries. In our study, participants had to perform the interaction tasks using both controllers. More specifically, they used the trigger key to start interaction events. In the virtual environment, each controller emits a pink light beam. In order to select a puzzle piece, participants had to project the light ray onto the piece, then press and hold the trigger key. To move a piece, they had to select the piece, and while holding the trigger, move the beam to a desired position. The rotating and scaling actions required similar operations, but with both controllers at the same time. Participants had to press both trigger keys and move the controllers closer together to scale down the selected piece or move the controllers further apart to scale the piece up. For rotating a selected piece, participants had to hold both trigger keys and move one controller around the other. The distance and the angle between two fingers when using *VXSlate* corresponds to the distance and the angle between two touch controllers. The distance and the angle can change simultaneously, thus scaling and rotating the selected piece. Although each touch controller has a joystick which can be used to perform translation, rotation, or scaling manipulations, we chose interaction techniques based on movements of the users' hands. These movements have been shown to be natural and intuitive to users in similar content manipulation tasks [59]. However, rotation and scaling manipulations cannot be performed on a piece being moved; moving manipulation can be done when there is only one finger (using *VXSlate*) or one touch controller interacting with the piece.

4.3 Participants

There were 17 participants recruited for the user study: ten female and seven male. Their ages varied from 19 to 41 (M = 25.3, SD = 6.96). Participants were students or academic staff members. All participants had normal or corrected-to-normal vision. Three were left-handed. None reported issues with balance. One reported a minor damage of her right side of bony labyrinth (that does not affect current abilities).

Participants had diverse backgrounds ranging from military technology development, software testing, and academia, to undergraduate students in social sciences and humanities (a dominant subgroup of 13 out of 17). Twelve out of 17 had never played a console game, nine reported never playing games on a PC, and only four declared playing on a PC

or smartphone several times per week. Although every participant declared using smartphones every day, 12 had never used a tablet. Additionally, the group was also inexperienced with any other touchscreen devices: nine never having used any such devices; five declaring an occasional use of once a month, and only three a weekly use. The participants were also not familiar with either VR or AR: 15 had never played with VR or AR, and only two declared playing once a month.

4.4 Procedure

Once participants arrived, first they were required to put on protection gear (mask, disposable isolation gown) to prevent potential spreading of SARS-CoV-2. Next, they were asked to consent on their health status to comply with regulations related to SARS-CoV-2. If they did not have any potential health issues, they would be introduced to the user study and data handling policy. They were told that they could withdraw at any time. Following that, participants answered a consent form on their participation in the study and a demographic questionnaire providing their background and self-assessing their technological expertise. After that, they were randomly assigned to perform the study using one of the two interaction techniques, which they would use to perform both stages (*Matching stage* and *Sketching stage*) of the game. Participants were then introduced to the task they needed to complete in the user study, and invited to watch an instruction video which introduced the interaction technique they would use and the steps to complete the task. After watching the instruction video, participants had a training session to familiarize them with the interaction technique and the task, in which they played the game with a 2×2 puzzle which was simpler to solve in *Matching stage* and had an easier path for tracing in *Sketching stage* compared to the one in the user study task (see Figure 3b). This training task was used for both interaction techniques. After finishing the training task, participants could start the user study. They first performed the *Matching stage*. After completing the *Matching stage*, they were asked to take off the VR headset and answer a *post-task questionnaire* to evaluate their perceived workload and experience in the *Matching stage*. After completing the questionnaire, participants put on the VR headset and continued to perform the *Sketching stage* of the task. When the *Sketching stage* was completed, they answered another *post-task questionnaire* similar to that of the *Matching stage* to evaluate their perceived workload and experience in the *Sketching stage*. Upon completing this, they were asked to answer a *post-Study questionnaire* on the usability of the interaction technique and open-ended questions on its advantages and disadvantages. Each participant performed only one trial of the task. In total, it took each participant around one hour to complete the study.

4.5 Data collection and measures

To evaluate participants' performance, we measured their **task completion times (TCT)** for each stage of the game (measured in seconds). For the *Sketching stage*, as participants traced the predefined path in two turns, we also measured completion times for sketching in each turn. For the *Matching stage*, we were also interested in measuring how much time it took in total for each participant to perform different manipulations on puzzle pieces (moving, scaling, rotating). We believe that manipulation times can provide insights into the effects of *VXSlate* on different docking tasks.

In the *Sketching stage*, tracing paths were recorded as a collection of tracing points. For each tracing point, we recorded its location on the large virtual display and whether or not it overlapped predefined paths. To determine whether a tracing point overlapped, we checked to see if the pixel at the same location on the large virtual display was yellow ($(R,G,B) = (255, 255, 0)$). For the *Sketching stage*, we also measured **sketching accuracy**, defined as a ratio of tracing points overlapping the predefined path over the total number of tracing points.

The *post-task questionnaire*, which was administered after each stage of the game, consisted of a NASA Task Load Index (NASA TLX) questionnaire to measure participants' perceived workload [36] and Self-Assessment Manikin (SAM) scales [16] to assess participants' emotions while performing the task using their assigned interaction technique. The *post-Study questionnaire* consisted of a System Usability Scale (SUS) questionnaire to assess the interaction techniques' usability [18], a Simulator Sickness Questionnaire (SSQ) to evaluate perceived sickness caused by the interaction techniques [43] and a User Experience Questionnaire (UEQ) to examine participants' user experience [47]. To gather qualitative feedback from the participants, we asked the following questions:

- What are your general impressions of the interaction technique in the study?
- What are the advantages of the interaction technique in the study?
- What are the disadvantages of the interaction technique in the study?
- What caused difficulties in using the interaction technique when interacting in VR?
- How would you improve the system/interaction in VR to help you achieve the goals of each task?

Qualitative data was then discussed by two researchers in the project to agree on common feedback.

4.6 Results

From 17 participants, we had nine data entries for touch controllers and eight data entries for VXSlate. We removed the data of a participant using VXSlate as this person did not follow the user study instructions. In total, we had nine data entries for touch controllers and seven data entries for VXSlate for data analysis. All of the quantitative data was analyzed using ANOVA.

4.6.1 Objective measures. For the *Matching stage*, the analysis result showed no significant difference in the overall *TCT* between the interaction techniques ($F(1, 15) = 0.37, p = 0.55$). On average, VXSlate seemed to take less time for participants to complete the task than the touch controllers did (354.58 ± 147.33 s vs. 403.44 ± 166.84 s). In addition, we also observed no significant difference in the total times devoted to moving ($F(1, 15) = 0.08, p = 0.78$) and rotating ($F(1, 15) = 0.34, p = 0.57$) puzzle pieces using VXSlate and touch controllers. In contrast, the total time of scaling using VXSlate was significantly shorter than that using touch controllers ($F(1, 15) = 5.92, p = 0.03$). The post-hoc test with Tukey-Kramer adjustment showed that VXSlate took less time for participants to complete the scaling manipulations than touch controllers did (44.46 ± 21.82 s vs 94.93 ± 51.09 s). Figure 4a shows average total times participants spent on manipulation tasks using VXSlate and touch controllers.

For the *Sketching stage*, the experimental results analysis showed no significant difference in completion times between the two techniques ($F(1, 15) = 0.9292, p = 0.35$). Participants spent on average around 85.18 (± 40.88) seconds with touch controllers and 107.63 (± 52.96) seconds with VXSlate for the task. We did not also observe significant difference in the participants' **sketching accuracy** between the two interaction techniques ($F(1, 15) = 2.45, p = 0.14$). The sketching accuracies were 92.32 (± 6.16)% for touch controllers and 87.40 (± 6.37)% for VXSlate. However, there was a difference in direction of the correlation between the completion time and sketching accuracy of the two interaction techniques. Correlation tests showed a positive weak correlation for touch controllers ($r(8) = 0.17, p = 0.67$), and a negative weak correlation for VXSlate ($r(6) = -0.08, p = 0.87$).

We also looked further into participants' performance in each sketching turn of the *Sketching stage*. For the first sketching turn, the result showed that there was no significant difference in the completion times between VXSlate and touch controllers (57.09 ± 35.96 s vs 37.44 ± 21.60 s, $F(1, 15) = 1.85, p = 0.19$) (see Figure 4b). In addition, we observed that there was also no significant difference between touch controllers and VXSlate in **sketching accuracy** (91.41 ± 8.06 %

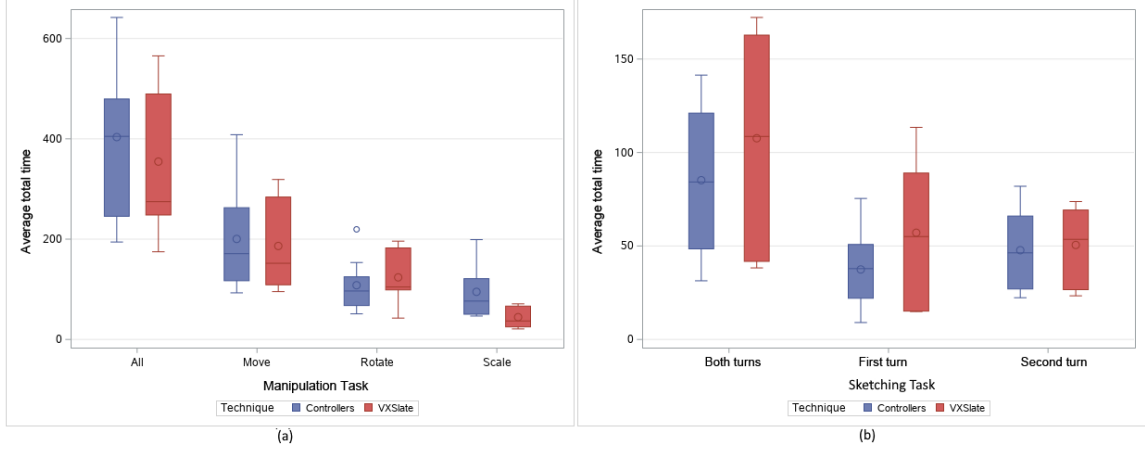


Fig. 4. (a) Average total times spent on manipulation tasks (all manipulations, move, rotate, scale) with puzzle pieces in the *Matching stage*, (b) Average completion times of the sketching task: entire task (both turns), the first turn and the second turn of sketching of *Sketching stage* using *touch controllers* and *VXSlate*.

vs $87.61 \pm 4.68\%$, $F(1, 15) = 1.22$, $p = 0.29$). Both touch controllers and *VXSlate* had weakly positive correlations between *sketching accuracy* and sketching times ($r(8) = 0.06$, $p = 0.89$ and $r(6) = 0.08$, $p = 0.87$).

Similar to the first sketch, we also observed no significant difference in the completion time between *VXSlate* and touch controllers (50.54 ± 19.51 s vs 47.75 ± 20.97 s, $F(1, 15) = 0.07$, $p = 0.79$) (see Figure 4b). Moreover, it was observed that there was also no significant difference between touch controllers and *VXSlate* in *sketching accuracy* ($93.24 \pm 4.20\%$ vs $87.02 \pm 9.18\%$, $F(1, 15) = 3.30$, $p = .09$). Both touch controllers and *VXSlate* had weakly positive correlations between *sketching accuracy* and sketching times ($r(8) = 0.35$, $p = 0.36$ and $r(6) = 0.04$, $p = 0.94$).

4.6.2 Subjective measures. In order to evaluate the subjective evaluation of mental demand, physical demand, temporal demand, effort, frustration, and performance, NASA TLX scores were used in a series of 2×2 mixed-design analyses of variance (ANOVAs). The interaction technique (*VXSlate* vs. touch controllers) was used in the analyses as a between-subject factor and the Task (*Matching stage* vs. *Sketching stage*) as a within-subject factor. The similarly designed ANOVAs were performed for three scales of the emotional aspect of subjective evaluation measured with the SAM scales (happiness, arousal, and control).

The analysis on perceived effort, the measure we chose for fatigue, revealed a significant interaction effect of the interaction technique and the task ($F(1, 15) = 9.04$, $p = 0.009$, $\eta^2 = 0.063$, see Figure 5). Decomposition of the interaction effect into pairwise comparisons with the HSD Tukey correction for multiple tests showed that the subjective effort significantly, $t(14) = 2.74$, $p = 0.016$, dropped in the *Sketching stage* ($M = 5.70$, $SE = 0.72$) compared to the *Matching stage* ($M = 6.85$, $SE = 0.72$) when participants were using *VXSlate*. This was in contrast to the condition where participants were using touch controllers, where the difference between *Matching stage* and *Sketching stage*, $t(14) = 1.21$, $p = 0.247$, was not significant.

We did not find any main effect of the interaction technique or the interaction effects on physical demand, mental demand, temporal demand, perceived frustration, or performance. Contrary to expectations, the analyses on the perceived emotional aspects of interaction in the experimental tasks with two different interaction techniques showed no significant effect for happiness, arousal, or control. We also performed Welch's two sample t-tests for comparing

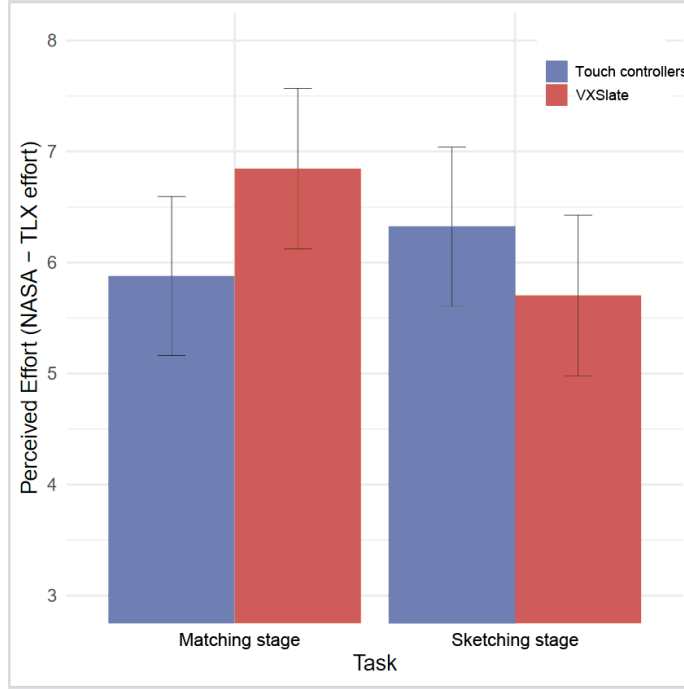


Fig. 5. Perceived effort (fatigue) interaction effect of Task and Controller. Bar height represents estimated means and error bars represent $\pm SE$ of the means.

general perception of the task usability (with the SUS questionnaire) and the user experience (with the UEQ questionnaire) between two conditions, *VXSlate* and touch controllers. The analysis on the system usability perception showed that the touch controller condition received a significantly higher average SUS score ($M = 3.94$) than *VXSlate* ($M = 3.40$), $t(21) = 2.74, p = 0.012$.

5 DISCUSSION

From the experiment analyses, the performance of *VXSlate* was comparable to the performance of touch controllers. For completing the *Matching stage*, even though there was no significant difference, *VXSlate* seemed to help participants take less time than touch controllers to perform docking tasks. With scaling, a docking task with a medium level of difficulty, *VXSlate* took a significantly shorter time than touch controllers did. Participants' feedback also indicated that *VXSlate* was more efficient for them in manipulation tasks in the *Matching stage* than in the *Sketching stage* (*Better control in the puzzle game (P14, using VXSlate)*). We argue that the passive haptic feedback of the physical tablet combined with a two-finger pinching gesture familiar to users helped participants scale objects more efficiently using *VXSlate* (*"Easy to use, allows intuitive movements and interactions" (P15, using VXSlate)*). In addition, the combination of head movements and multitouch interactions also provided users richer and more efficient ways to manipulate virtual contents (*"More possibilities to manipulate in VR. The operation with head movement was fast" (P13, using VXSlate)*).

For the sketching task, it took almost the same time for both techniques, with *VXSlate* taking a little longer. This was caused by the *virtual tablet* having smaller coverage than the large virtual display. While it might still help participants

sketch quite precisely, it required more time to navigate the *virtual tablet* along the predefined path to finish the tracing. Although no significant difference in *sketching accuracy* was found, statistical figures showed that participants sketched slightly less accurately using *VXSlate* compared to using touch controllers. Based on our observation, we suspect that this could be caused by certain technical limitations of the current *VXSlate* prototype. The first reason might be due to the so-called fat-finger problem [87] of the user hand image overlaid on the *virtual tablet*. The image of the participants' fingers was larger than the thickness of the predefined path in the study task, thus causing difficulties for the participants to correctly align their finger within the path (*"The shape of the fingers could be smaller. This would allow me to operate more accurately"* (P15, using *VXSlate*)). While imprecision in touch input is well understood for cases where the users can see their fingers directly (e.g. using a tablet) [39], the so-called fat-finger problem remains to be solved for mediated representations of fingers (e.g. combining tablet with VR). This issue can be mitigated in the future by using computer vision advancements to detect the user hand posture [65, 91] and stylize the visualization of the hand [44, 94] to reduce obtrusion, rather than showing the real images as in the current prototype. The second reason is the limited update frame rate of the user hand images compared to the update frequency of touch input. While touch input on the tablet was updated at 60 fps, we could only stream the user hand images from the webcam at 20 fps. The disparity between the user hand images and the captured touch input might cause difficulties for participants to synchronize the hand images with the touch position, thus leading to misalignment on the predefined path. We believe that if future systems use a camera with higher frame rates and utilizing faster network connection, this problem can be mitigated. As the current *VXSlate* is just a proof-of-concept, we believe that future implementations of this interaction technique using emerging technology advancements would deliver more promising user performance in highly precise tasks.

The lower SUS and user experience scores of *VXSlate* compared to touch controllers can be explained by the technology novelty effect. In fact, compared to well-designed and well-engineered commercial touch controllers, the setup of *VXSlate*, as a prototype, still appears to be bulkier and less user friendly (*"I was expecting a more mature system."* (P17, using *VXSlate*)) (P02, using *VXSlate*). The presence of the purple circle representing the user's current head direction was perceived to be distracting, especially when performing fine-tuning content manipulations (*"The purple dot was annoying while I was sketching. Can I remove or minimize it in my field of view?"* (P02, using *VXSlate*)). To improve this, we will apply a dynamic visualization approach, in which the circle will only be visible if the user's intentional head movements are detected. It will fade away if the user's head movements are considered unintentional. Another issue contributing to the reduced perceived usability of *VXSlate* was that sometimes the *virtual tablet* did not move with the participant's head direction as expected (*"In spite of me moving my head, sometimes the virtual tablet was not changing"* (P02, using *VXSlate*)). We observed two causes of this problem. First, the technique that we used to detect intentional head movements failed in some cases. Further improvements will be needed to make the algorithm more robust. Second, sometimes the thumb of the participant's hand holding the tablet unintentionally touched the screen while he/she moved their head. If he/she unconsciously moved his/her touch point out of the screen, the *virtual tablet* surprisingly shifted to a new position. In the future, this might be avoided by using animations of the user's touch points that are shown on the virtual tablet when the system detects new intentional head movements. Such a solution might help the user become aware of unintentional touches on the tablet's screen. Even though participants were shown how to scale the *virtual tablet* up and down, they did not always remember this feature (*"Making the tablet as twice as big."* (P14, using *VXSlate*)); adding a widget (e.g. a magnifier icon) at a corner of the *virtual tablet* would remind the users of this feature.

Finally, more important seems to be the result on perceived effort during task completion. The analyses clearly showed that the perceived effort dropped in the second task when using *VXSlate* and this does not change when using touch controller. This is particularly noticeable because the second manipulation task (sketching) was considered to be even more difficult than the manipulation tasks in the first stage. In fact, this higher level of difficulty was reflected in the touch-controller condition, where the participants reported spending more effort in the second stage than in the first (*"Drawing was difficult"* (P06, using touch controllers)). Interactions with body gestures while using touch controllers were also reported to cause fatigue (*"The interaction with the body gestures was tiring; you need more training to learn it."* (P04, using touch controllers)) In contrast, the participants' perceived effort using *VXSlate* did not increase but even decreased even as the manipulation became more difficult. We thus argue that when users get more familiar with *VXSlate*, they will interact with VR more effortlessly, and one may expect that it will also influence the general evaluation of the usability and user experience. Such a conclusion, although justified from the obtained results, needs additional confirmation in future longitudinal studies evaluating explicit learning effects of an interaction technique. Besides that, the fact that users do not need to spend effort on body gestures or physically navigating around in the virtual environment when using *VXSlate* was also favored by participants (*"Good coordination of the gaze cursor gives me a feeling of 'being static in the space'"* (P15, using *VXSlate*)). In a broader perspective, this advantage of *VXSlate* can make it potentially contribute to assistive interactive technology. For example, making use of *VXSlate* interaction technique, we could enable people with mobility disability to work in or experience VR without having to physically navigate around the environment.

6 POTENTIAL APPLICATIONS OF VXSlate

VXSlate has the potential to support users to work with VR large displays and high-resolution contents in different domains, such as medical imaging diagnosis, traffic management or emergency response. We also believe the *VXSlate* design space can be explored for mixed reality (MR). For both VR and MR, controlling a virtual display whose size is being changed dynamically with the handheld physical surface, *VXSlate*'s form factor can enable fluent task-switching among different screens. Also, the ability to work with a large display anywhere at any time using a compact hardware setup provided by *VXSlate* opens the opportunities for collaborating remotely on large displays in novel team settings, which could also fully integrate mobile workers. In this section, we also demonstrate these possibilities using two applications. The first application is a mixed-reality sketchpad that allows a user to flexibly extend the screen space of a tablet for efficient drawing. The second application is an interface that allows a remote person to effectively collaborate with their team who are working with a large display. This application is especially useful for traveling individuals who prefer using compact computer hardware setups (e.g. tablet, smartphone or laptop), which are as yet not well supported in remote collaboration involving large displays.

6.1 MR extendable sketchpad

Designers and artists often create their paintings, graphics, and designs using tablet-sized sketchpads. While drawing on such devices, the user frequently needs to zoom in and out in order to switch between the overview of the entire graphical work and details of a specific area. Drawing-interfaces typically provide a thumbnail sketch which shows the graphical the and the current location of the focused area. Artists still frequently zoom in and out in order to switch between the overall and the focus view to see the entire work at a higher resolution. Also, if the artist wants to navigate from one focused area to another, he/she needs to zoom out, navigate, and zoom in again.

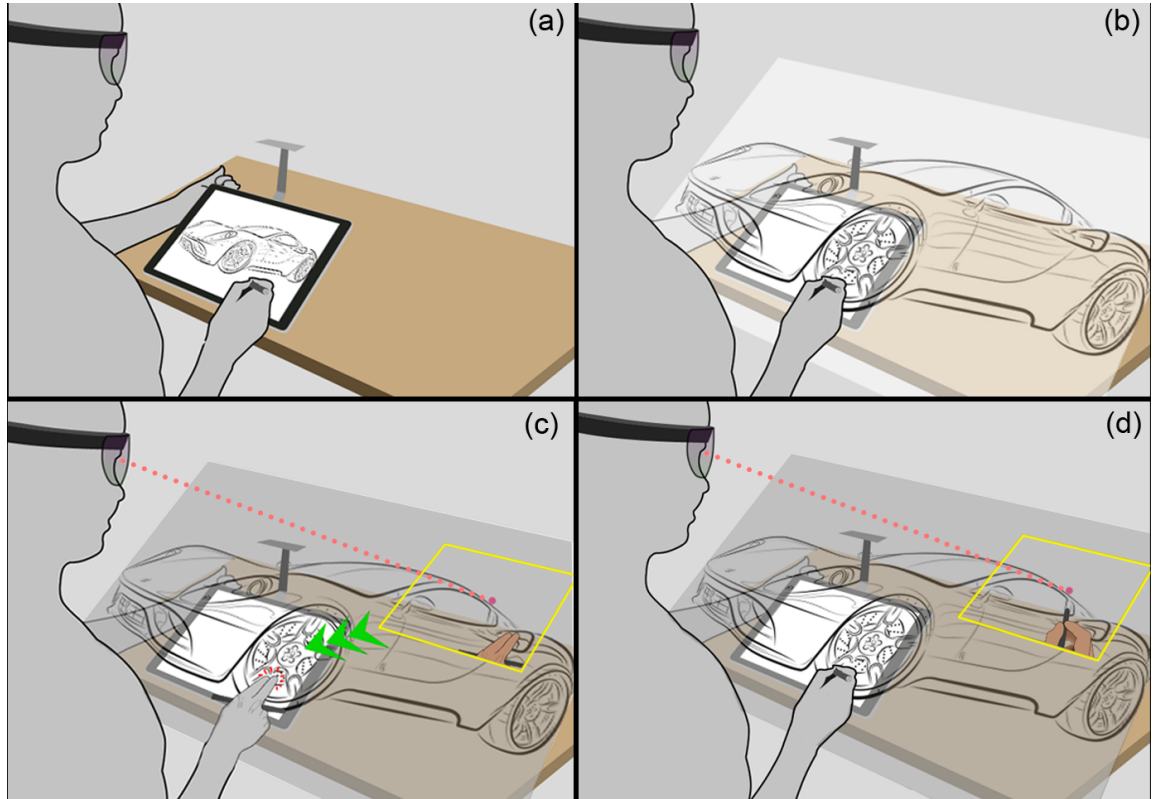


Fig. 6. Mixed-reality extendable sketchpad concept where the user (a) sketches on a tablet as usual (b) sees the entire drawing extended outside the tablet when zooming into a detail (c) can quickly change the focused area for drawing by using facing direction and a double-tap on the tablet (d) distally draws on an area outside the tablet.

VXSlate can be employed to reduce view switching when drawing on tablet-sized devices. The user creates a drawing using touch interactions on the tablet as usual (see Figure 6a). When zooming into a specific area to create more details, the focused area will be shown on the tablet while the rest of the graphical work will be enlarged and extended beyond the tablet and can be seen through the MR headset (see Figure 6b). This virtual high-resolution representation allows the user to perform fine-tuned drawing without losing their overview on how the local modifications on the focused area will be integrated into the whole picture of the graphical work. Besides that, the user can also quickly change the focused area to work on the artwork by simply gazing at it and confirming the selection using certain touch gestures on the tablet (e.g. double-tap using two fingers as shown in Figure 6c). Once such a selection is performed, the entire graphical work will be shifted so that the selected area is aligned on the tablet to help the user draw using touch direction. The *VXSlate* concept can also introduce a remote sketching feature for this application. The user can gaze at an area not on the tablet where they want to draw, and a virtual representation of the tablet screen and the images of their hands on and above the screen will be shown there. Next, the user can remotely align his/her hand at the intended position and perform the drawing (see Figure 6d). This feature could be useful when the user just wants to change some small details on a far-off area on the graphical work, but does not wish to shift that area to the tablet. These features

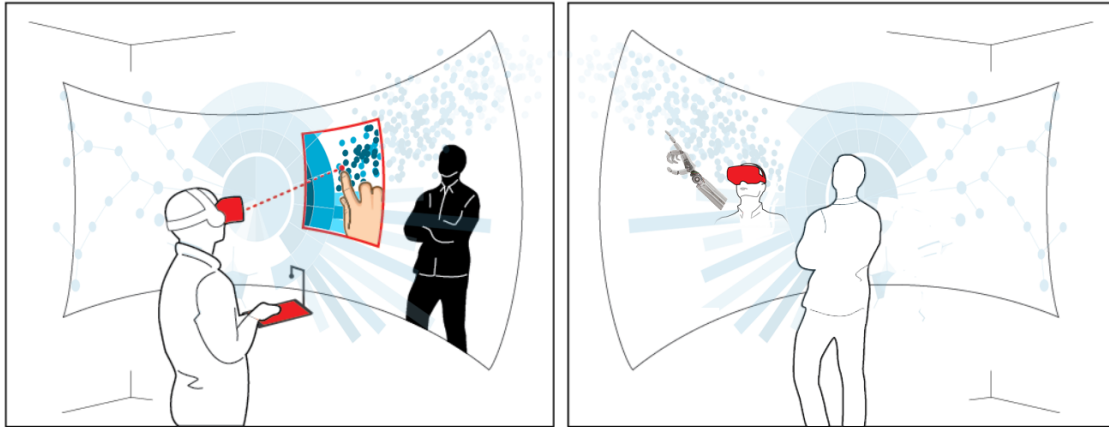


Fig. 7. Concept of using VXSlate for remote collaboration with physical large displays. Left: a traveling individual remotely working with a large virtual display using VXSlate. Right: A remote collaborator working with a large physical display can perceive deictic gestures and head movements of the remote person captured by the VXSlate system.

together allow the user to perform drawing conveniently on high-resolution contents using a compact hardware setup that consumes even less space than a professional sketchpad.

6.2 VR large display remote collaboration

Collaborating remotely on large displays has been investigated in several problem-solving scenarios [45, 52, 53]. However, existing systems assume a symmetric hardware setup where large displays are available on all collaboration endpoints. Remote individuals, for example those who are traveling and using personal devices like tablets, still remain unsupported to remotely collaborate with teams working with large displays. This leads to several location constraints for team collaboration using large displays. In fact, there are numerous challenges to support asymmetric remote collaboration setups between tablets and large displays. First, mapping from touch interactions on a tablet to a large display can lead to inaccurate manipulations, causing confusion and frustration among collaborators. Second, non-verbal communication involved in the collaboration might be misinterpreted. In symmetric setups, non-verbal communication of collaborators, such as gaze or deictic gestures can be easily interpreted correctly thanks to similar physical arrangements among the partners [6, 7]. Nevertheless, in asymmetric setups consisting of a tablet and a large display, these cues can be misinterpreted due to different spatiality between two sides.

VXSlate can be an approach to deal with the aforementioned issues. First, it provides a way for the user to efficiently perform fine-tuned content manipulations on large virtual displays by combining facing direction and touch interaction on a handheld tablet. Second, employing hardware, like MirrorTablet [48], VXSlate can capture both hand gestures and facing direction through the tablet's front-facing camera. These communication cues can be combined to convey the user's attention and interaction on the large physical display of the remote sides (see Figure 7). The position of the video feed of the user's head movements on the large physical display can be determined based on the relative location of the user to the large virtual display. In Figure 7, we propose an approach to visually connect the video feed of the user's head movements with stylized hands (supported by hand tracking advancements [91]) using a graphical robot arm. This approach can provide a simple humanoid representation to connect the two non-verbal cues (pointing &

head orientation) while still avoiding the uncanny valley problem [74]. Altogether, *VXSlate* opens the opportunity for traveling individuals using personal devices such as a tablet to collaborate efficiently with remote collaborators using large physical displays.

7 CONCLUSION AND OUTLOOK

In this work, we have proposed and explored the design of *VXSlate*; a novel VR concept incorporating two input streams—head movement and touch tablet—into a non-traditional combination. The *VXSlate* aims for a portable setup using commercial devices only. Head movements are tracked by a VR headset while touch input takes place on a handheld tablet. We demonstrate how this device combination provides a powerful way to interact with nearby large virtual displays. *VXSlate* allows its users to perform a wide range of content manipulations at varying precision levels and ranging from selecting, to docking (moving, scaling, rotating, positioning), and to steering.

Our user study comparing *VXSlate* with off-the-shelf touch-controller interactions showed that in general, *VXSlate* is comparable to commercial touch controllers in supporting participants to perform tasks at different levels of difficulty. However, for scaling (a content manipulation task with a medium level of difficulty), *VXSlate* significantly reduced participants' completion times when compared to completion times with touch-controller interactions. It is also important that after an interval of interaction, participants reported lower effort (fatigue) in using *VXSlate* when compared to users of touch controllers. Taking into account the technology novelty effect, this might suggest the potential advantage of *VXSlate* over touch controllers.

The design space of *VXSlate* can also be further explored in different contexts, such as mixed reality, remote collaboration with asymmetric settings, and even assistive technology for people with mobility disability. We believe that these and other future applications of *VXSlate* could stimulate research into the input duet proposed here—head movement and mobile touch—for large virtual display interaction.

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REFERENCES

- [1] Johnny Accot and Shumin Zhai. 1999. Performance evaluation of input devices in trajectory-based tasks: an application of the steering law. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 466–472. <https://doi.org/10.1145/302979.303133>
- [2] Christopher Andrews, Alex Endert, Beth Yost, and Chris North. 2011. Information visualization on large, high-resolution displays: Issues, challenges, and opportunities. *Information Visualization* 10, 4 (2011), 341–355. <https://doi.org/10.1177/1473871611415997>
- [3] Rahul Arora, Rubaiat Habib Kazi, Tovi Grossman, George Fitzmaurice, and Karan Singh. 2018. *SymbiosisSketch: Combining 2D & 3D Sketching for Designing Detailed 3D Objects in Situ*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3173574.3173759>
- [4] Rahul Arora, Rubaiat Habib Kazi, Fraser Anderson, Tovi Grossman, Karan Singh, and George W Fitzmaurice. 2017. Experimental Evaluation of Sketching on Surfaces in VR. In *CHI*, Vol. 17. ACM, New York, NY, USA, 5643–5654. <https://doi.org/10.1145/3025453.3025474>
- [5] Rowel Atienza, Ryan Blonna, Maria Isabel Saldares, Joel Casimiro, and Vivencio Fuentes. 2016. Interaction techniques using head gaze for virtual reality. In *2016 IEEE Region 10 Symposium (TENSYP)*. IEEE, New York, NY, USA, 110–114. <https://doi.org/10.1109/TENCONSpring.2016.7519387>

- [6] Ignacio Avellino, Cédric Fleury, and Michel Beaudouin-Lafon. 2015. Accuracy of deictic gestures to support telepresence on wall-sized displays. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 2393–2396. <https://doi.org/10.1145/2702123.2702448>
- [7] Ignacio Avellino, Cédric Fleury, Wendy E. Mackay, and Michel Beaudouin-Lafon. 2017. CamRay: Camera Arrays Support Remote Collaboration on Wall-Sized Displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 6718–6729. <https://doi.org/10.1145/3025453.3025604>
- [8] Liz Bacon, L MacKinnon, Amedeo Cesta, and Gabriella Cortellessa. 2013. Developing a smart environment for crisis management training. *Journal of Ambient Intelligence and Humanized Computing* 4, 5 (2013), 581–590. <https://doi.org/10.1007/s12652-012-0124-0>
- [9] Ravin Balakrishnan and Gordon Kurtenbach. 1999. Exploring bimanual camera control and object manipulation in 3D graphics interfaces. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 56–62. <https://doi.org/10.1145/302979.302991>
- [10] Amartya Banerjee, Jesse Burstyn, Audrey Girouard, and Roel Vertegaal. 2012. MultiPoint: Comparing laser and manual pointing as remote input in large display interactions. *International Journal of Human-Computer Studies* 70, 10 (2012), 690–702. <https://doi.org/10.1016/j.ijhcs.2012.05.009>
- [11] Jonas Blattgerste, Patrick Renner, and Thies Pfeiffer. 2018. Advantages of eye-gaze over head-gaze-based selection in virtual and augmented reality under varying field of views. In *Proceedings of the Workshop on Communication by Gaze Interaction*. ACM, New York, NY, USA, 1–9. <https://doi.org/10.1145/3206343.3206349>
- [12] Sebastian Boring, Dominikus Baur, Andreas Butz, Sean Gustafson, and Patrick Baudisch. 2010. Touch projector: mobile interaction through video. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 2287–2296. <https://doi.org/10.1145/1753326.1753671>
- [13] Sebastian Boring, Dominikus Baur, Andreas Butz, Sean Gustafson, and Patrick Baudisch. 2010. Touch projector: mobile interaction through video. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 2287–2296. <https://doi.org/10.1145/1753326.1753671>
- [14] Sebastian Boring, Marko Jurmu, and Andreas Butz. 2009. Scroll, tilt or move it: using mobile phones to continuously control pointers on large public displays. In *Proceedings of the 21st Annual Conference of the Australian Computer-Human Interaction Special Interest Group: Design: Open 24/7*. ACM, New York, NY, USA, 161–168. <https://doi.org/10.1145/1738826.1738853>
- [15] Sebastian Boring, Marko Jurmu, and Andreas Butz. 2009. Scroll, tilt or move it: using mobile phones to continuously control pointers on large public displays. In *Proceedings of the 21st Annual Conference of the Australian Computer-Human Interaction Special Interest Group: Design: Open 24/7*. ACM, New York, NY, USA, 161–168. <https://doi.org/10.1145/1738826.1738853>
- [16] Margaret M Bradley and Peter J Lang. 1994. Measuring emotion: the self-assessment manikin and the semantic differential. *Journal of behavior therapy and experimental psychiatry* 25, 1 (1994), 49–59. [https://doi.org/10.1016/0005-7916\(94\)90063-9](https://doi.org/10.1016/0005-7916(94)90063-9)
- [17] Eugénie Brasier, Olivier Chapuis, Nicolas Ferey, Jeanne Vezien, and Caroline Appert. 2020. ARPADs: Mid-air Indirect Input for Augmented Reality. In *2020 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, New York, NY, USA, 332–343.
- [18] JSUS Brooke. 1996. *A quick and dirty usability scale*. *Usability evaluation in industry*. Taylor & Francis Ltd, London, UK, Chapter 21, 189–194.
- [19] William Buxton, George Fitzmaurice, Ravin Balakrishnan, and Gordon Kurtenbach. 2000. Large displays in automotive design. *IEEE Computer Graphics and applications* 20, 4 (2000), 68–75. <https://doi.org/10.1109/38.851753>
- [20] Qi Chen, John Grundy, and John Hosking. 2003. An e-whiteboard application to support early design-stage sketching of UML diagrams. In *IEEE Symposium on Human Centric Computing Languages and Environments, 2003. Proceedings. 2003*. IEEE, New York, NY, USA, 219–226. <https://doi.org/10.1109/HCC.2003.1260232>
- [21] Kelvin Cheng, Jane Li, and Christian Müller-Tomfelde. 2012. Supporting interaction and collaboration on large displays using tablet devices. In *Proceedings of the International Working Conference on Advanced Visual Interfaces*. ACM, New York, NY, USA, 774–775. <https://doi.org/10.1145/2254556.2254708>
- [22] Lung-Pan Cheng, Eyal Ofek, Christian Holz, Hrvoje Benko, and Andrew D Wilson. 2017. Sparse haptic proxy: Touch feedback in virtual environments using a general passive prop. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 3718–3728. <https://doi.org/10.1145/3025453.3025753>
- [23] Apoorve Chokshi, Teddy Seyed, Francisco Marinho Rodrigues, and Frank Maurer. 2014. ePlan multi-surface: A multi-surface environment for emergency response planning exercises. In *Proceedings of the Ninth ACM International Conference on Interactive Tabletops and Surfaces*. ACM, New York, NY, USA, 219–228. <https://doi.org/10.1145/2669485.2669520>
- [24] Nancy Dorigi and Barry Sullivan. 2003. FutureFlight Central: a revolutionary air traffic control tower simulation facility. In *AIAA Modeling and Simulation Technologies Conference and Exhibit*. American Institute of Aeronautics and Astronautics, Reston, VA, USA, 5598. <https://doi.org/10.2514/6.2003-5598>
- [25] Tobias Drey, Jan Gugenheimer, Julian Karlbauer, Maximilian Milo, and Enrico Rukzio. 2020. VRSketchIn: Exploring the Design Space of Pen and Tablet Interaction for 3D Sketching in Virtual Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376628>
- [26] Navid Farahani, Robert Post, Jon Duboy, Ishtiaque Ahmed, Brian J Kolowitz, Teppituk Krinchai, Sara E Monaco, Jeffrey L Fine, Douglas J Hartman, and Liron Pantanowitz. 2016. Exploring virtual reality technology and the Oculus Rift for the examination of digital pathology slides. *Journal of pathology informatics* 7 (2016), 22. <https://doi.org/10.4103/2153-3539.181766>

- [27] Jamie Feltham. 2017. CollabHub Is A Virtual Whiteboard With Plenty Of Possibilities. <https://uploadvr.com/collabhub-virtual-whiteboard-plenty-possibilities/>.
- [28] Tiare Feuchtnr and Jörg Müller. 2018. Ownershift: Facilitating Overhead Interaction in Virtual Reality with an Ownership-Preserving Hand Space Shift. In *The 31st Annual ACM Symposium on User Interface Software and Technology*. ACM, New York, NY, USA, 31–43. <https://doi.org/10.1145/3242587.3242594>
- [29] Alexandra Frischen, Andrew P Bayliss, and Steven P Tipper. 2007. Gaze cueing of attention: visual attention, social cognition, and individual differences. *Psychological bulletin* 133, 4 (2007), 694. <https://doi.org/10.1037/0033-2909.133.4.694>.
- [30] Emma Gowen and R Chris Miall. 2006. Eye–hand interactions in tracing and drawing tasks. *Human movement science* 25, 4-5 (2006), 568–585. <https://doi.org/10.1016/j.humov.2006.06.005>
- [31] Jan Gugenheimer, Evgeny Stemasov, Harpreet Sareen, and Enrico Rukzio. 2018. FaceDisplay: Towards Asymmetric Multi-User Interaction for Nomadic Virtual Reality. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3173628>
- [32] Daniel Guitton and Michel Volle. 1987. Gaze control in humans: eye-head coordination during orienting movements to targets within and beyond the oculomotor range. *Journal of neurophysiology* 58, 3 (1987), 427–459. <https://doi.org/10.1152/jn.1987.58.3.427>
- [33] Michael Haller, Jakob Leitner, Thomas Seifried, James R Wallace, Stacey D Scott, Christoph Richter, Peter Brandl, Adam Gokceade, and Seth Hunter. 2010. The nice discussion room: Integrating paper and digital media to support co-located group meetings. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 609–618. <https://doi.org/10.1145/1753326.1753418>
- [34] Robert Hardy and Enrico Rukzio. 2008. Touch & interact: touch-based interaction of mobile phones with displays. In *Proceedings of the 10th international conference on Human computer interaction with mobile devices and services*. ACM, New York, NY, USA, 245–254. <https://doi.org/10.1145/1409240.1409267>
- [35] Beverly L Harrison, Gordon Kurtenbach, and Kim J Vicente. 1995. An experimental evaluation of transparent user interface tools and information content. In *ACM Symposium on User Interface Software and Technology*. ACM, New York, NY, USA, 81–90. <https://doi.org/10.1145/215585.215669>
- [36] Sandra G Hart and Lowell E Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. In *Advances in psychology*. Vol. 52. Elsevier, London, UK, 139–183. [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)
- [37] Luke Hespanhol, Martin Tomitsch, Kazjon Grace, Anthony Collins, and Judy Kay. 2012. Investigating intuitiveness and effectiveness of gestures for free spatial interaction with large displays. In *Proceedings of the 2012 International Symposium on Pervasive Displays*. ACM, New York, NY, USA, 6. <https://doi.org/10.1145/2307798.2307804>
- [38] Jari K Hietanen. 1999. Does your gaze direction and head orientation shift my visual attention? *Neuroreport* 10, 16 (1999), 3443–3447. <https://doi.org/10.1097/00001756-199911080-00033>
- [39] Christian Holz and Patrick Baudisch. 2010. The generalized perceived input point model and how to double touch accuracy by extracting fingerprints. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACN, New York, NY, USA, 581–590.
- [40] Hikaru Ibayashi, Yuta Sugiura, Daisuke Sakamoto, Natsuki Miyata, Mitsunori Tada, Takashi Okuma, Takeshi Kurata, Masaaki Mochimaru, and Takeo Igarashi. 2015. Dollhouse VR: A Multi-View, Multi-User Collaborative Design Workspace with VR Technology. In *SIGGRAPH Asia 2015 Posters* (Kobe, Japan) (SA '15). Association for Computing Machinery, New York, NY, USA, Article 24, 1 pages. <https://doi.org/10.1145/2820926.2820948>
- [41] Petra Isenberg, Pierre Dragicevic, Wesley Willett, Anastasia Bezerianos, and Jean-Daniel Fekete. 2013. Hybrid-image visualization for large viewing environments. *IEEE transactions on visualization and computer graphics* 19, 12 (2013), 2346–2355.
- [42] Ricardo Jota, Miguel A Nacenta, Joaquim A Jorge, Sheelagh Carpendale, and Saul Greenberg. 2010. A comparison of ray pointing techniques for very large displays. In *Proceedings of graphics interface 2010*. Canadian Information Processing Society, Mississauga, ON, Canada, 269–276. <https://doi.org/10.11575/PRISM/35537>
- [43] Robert S Kennedy, Norman E Lane, Kevin S Berbaum, and Michael G Lilienthal. 1993. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The international journal of aviation psychology* 3, 3 (1993), 203–220. https://doi.org/10.1207/s15327108ijap0303_3
- [44] Chelhwon Kim, Patrick Chiu, Joseph de la Pena, Laurent Denoue, Jun Shingu, and Yulius Tjahjadi. 2019. Toward Long Distance Tabletop Hand-Document Telepresence. In *Proceedings of the 2019 ACM International Conference on Interactive Surfaces and Spaces*. ACM, New York, NY, USA, 391–396.
- [45] Scott R Klemmer, Mark W Newman, Ryan Farrell, Mark Bilezikjian, and James A Landay. 2001. The designers’ outpost: a tangible interface for collaborative web site. In *Proceedings of the 14th annual ACM symposium on User interface software and technology*. ACM, New York, NY, USA, 1–10. <https://doi.org/10.1145/502348.502350>
- [46] Mikko Kytö, Barrett Ens, Thammathip Piumsomboon, Gun A Lee, and Mark Billinghurst. 2018. Pinpointing: Precise Head-and-Eye-Based Target Selection for Augmented Reality. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 81. <https://doi.org/10.1145/3173574.3173655>
- [47] Bettina Laugwitz, Theo Held, and Martin Schrepp. 2008. Construction and evaluation of a user experience questionnaire. In *Symposium of the Austrian HCI and usability engineering group*. Springer, Berlin, Germany, 63–76. https://doi.org/10.1007/978-3-540-89350-9_6
- [48] Khanh-Duy Le, Kening Zhu, and Morten Fjeld. 2017. Mirrortablet: exploring a low-cost mobile system for capturing unmediated hand gestures in remote collaboration. In *Proceedings of the 16th International Conference on Mobile and Ubiquitous Multimedia*. ACM, New York, NY, USA, 79–89. <https://doi.org/10.1145/3152832.3152838>

- [49] Robert W Lindeman, John L Sibert, and James K Hahn. 1999. Hand-held windows: towards effective 2D interaction in immersive virtual environments. In *Proceedings IEEE Virtual Reality (Cat. No. 99CB36316)*. IEEE, New York, NY, USA, 205–212. <https://doi.org/10.1109/VR.1999.756952>
- [50] Can Liu, Olivier Chapuis, Michel Beaudouin-Lafon, and Eric Lecolinet. 2017. CoReach: Cooperative gestures for data manipulation on wall-sized displays. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 6730–6741. <https://doi.org/10.1145/3025453.3025594>
- [51] Shahzad Malik, Abhishek Ranjan, and Ravin Balakrishnan. 2005. Interacting with large displays from a distance with vision-tracked multi-finger gestural input. In *Proceedings of the 18th annual ACM symposium on User interface software and technology*. ACM, New York, NY, USA, 43–52. <https://doi.org/10.1145/1095034.1095042>
- [52] Thomas Marrinan, Jillian Aurisano, Arthur Nishimoto, Krishna Bharadwaj, Victor Mateevitsi, Luc Renambot, Lance Long, Andrew Johnson, and Jason Leigh. 2014. SAGE2: A new approach for data intensive collaboration using Scalable Resolution Shared Displays. In *10th IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing*. IEEE, New York, NY, USA, 177–186. <https://doi.org/10.4108/icst.collaboratecom.2014.257337>
- [53] Thomas Marrinan, Jason Leigh, Luc Renambot, Angus Forbes, Steve Jones, and Andrew E Johnson. 2017. Mixed presence collaboration using scalable visualizations in heterogeneous display spaces. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing*. ACM, New York, NY, USA, 2236–2245. <https://doi.org/10.1145/2998181.2998346>
- [54] David C McCallum and Pourang Irani. 2009. ARC-Pad: absolute+ relative cursor positioning for large displays with a mobile touchscreen. In *Proceedings of the 22nd annual ACM symposium on User interface software and technology*. ACM, New York, NY, USA, 153–156. <https://doi.org/10.1145/1622176.1622205>
- [55] R. A. Montano-Murillo, C. Nguyen, R. H. Kazi, S. Subramanian, S. DiVerdi, and D. Martinez-Plasencia. 2020. Slicing-Volume: Hybrid 3D/2D Multi-target Selection Technique for Dense Virtual Environments. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, New York, NY, USA, 53–62. <https://doi.org/10.1109/VR46266.2020.00023>
- [56] Brad A Myers, Rishi Bhatnagar, Jeffrey Nichols, Choon Hong Peck, Dave Kong, Robert Miller, and A Chris Long. 2002. Interacting at a distance: measuring the performance of laser pointers and other devices. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. ACM, New York, NY, USA, 33–40. <https://doi.org/10.1145/503376.503383>
- [57] Mathieu Nancel, Olivier Chapuis, Emmanuel Pietriga, Xing-Dong Yang, Pourang P Irani, and Michel Beaudouin-Lafon. 2013. High-precision pointing on large wall displays using small handheld devices. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 831–840. <https://doi.org/10.1145/2470654.2470773>
- [58] Mathieu Nancel, Emmanuel Pietriga, Olivier Chapuis, and Michel Beaudouin-Lafon. 2015. Mid-air pointing on ultra-walls. *ACM Transactions on Computer-Human Interaction (TOCHI)* 22, 5 (2015), 21. <https://doi.org/10.1145/2766448>
- [59] Vijayakumar Nanjappan, Hai-Ning Liang, Feiyu Lu, Konstantinos Papangelis, Yong Yue, and Ka Lok Man. 2018. User-elicited dual-hand interactions for manipulating 3D objects in virtual reality environments. *Human-centric Computing and Information Sciences* 8, 1 (2018), 1–16.
- [60] Toshikazu Ohshima, Hiroyuki Yamamoto, and Hideyuki Tamura. 1996. Gaze-directed adaptive rendering for interacting with virtual space. In *Proceedings of the IEEE 1996 Virtual Reality Annual International Symposium*. IEEE, New York, NY, USA, 103–110. <https://doi.org/10.1109/VRAIS.1996.490517>
- [61] Dan R Olsen Jr and Travis Nielsen. 2001. Laser pointer interaction. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. ACM, New York, NY, USA, 17–22. <https://doi.org/10.1145/365024.365030>
- [62] Florian Perteneder, Martin Bresler, Eva-Maria Grossauer, Joanne Leong, and Michael Haller. 2015. cluster: Smart clustering of free-hand sketches on large interactive surfaces. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology*. ACM, New York, NY, USA, 37–46. <https://doi.org/10.1145/2807442.2807455>
- [63] Ken Pfeuffer, Benedikt Mayer, Diako Mardanbegi, and Hans Gellersen. 2017. Gaze+ pinch interaction in virtual reality. In *Proceedings of the 5th Symposium on Spatial User Interaction*. ACM, New York, NY, USA, 99–108. <https://doi.org/10.1145/3131277.3132180>
- [64] Krzysztof Pietroszek, James R Wallace, and Edward Lank. 2015. Tiltcasting: 3d interaction on large displays using a mobile device. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology*. ACM, New York, NY, USA, 57–62. <https://doi.org/10.1145/2807442.2807471>
- [65] Pramod Kumar Pisharady, Prahlad Vadakkepat, and Ai Poh Loh. 2013. Attention based detection and recognition of hand postures against complex backgrounds. *International Journal of Computer Vision* 101, 3 (2013), 403–419.
- [66] Arnaud Prouzeau, Anastasia Bezerianos, and Olivier Chapuis. 2016. Towards road traffic management with forecasting on wall displays. In *Proceedings of the 2016 ACM International Conference on Interactive Surfaces and Spaces*. ACM, New York, NY, USA, 119–128. <https://doi.org/10.1145/2992154.2992158>
- [67] Yuan Yuan Qian and Robert J Teather. 2017. The eyes don't have it: an empirical comparison of head-based and eye-based selection in virtual reality. In *Proceedings of the 5th Symposium on Spatial User Interaction*. ACM, New York, NY, USA, 91–98. <https://doi.org/10.1145/3131277.3132182>
- [68] Raf Ramakers, Davy Vanacken, Kris Luyten, Karin Coninx, and Johannes Schöning. 2012. Carpus: a non-intrusive user identification technique for interactive surfaces. In *Proceedings of the 25th annual ACM symposium on User interface software and technology*. ACM, New York, NY, USA, 35–44. <https://doi.org/10.1145/2380116.2380123>
- [69] Adrian Ramcharitar and Robert J Teather. 2018. EZCursorVR: 2D selection with virtual reality head-mounted displays. In *Proceedings of the 44th Graphics Interface Conference*. Canadian Human-Computer Communications Society, Mississauga, ON, Canada, 123–130. <https://doi.org/10.20380/GI2018.17>

- [70] M. Rauterberg, M. Bichsel, M. Meier, and M. Fjeld. 1997. A gesture based interaction technique for a planning tool for construction and design. In *Proceedings 6th IEEE International Workshop on Robot and Human Communication. RO-MAN'97 SENDAI*. IEEE, New York, NY, USA, 212–217. <https://doi.org/10.1109/ROMAN.1997.646984>
- [71] Valentin Schwind, Jens Reinhardt, Rufat Rzaev, Niels Henze, and Katrin Wolf. 2018. Virtual reality on the go?: a study on social acceptance of VR glasses. In *Proceedings of the 20th International Conference on Human-Computer Interaction with Mobile Devices and Services Adjunct*. ACM, New York, NY, USA, 111–118. <https://doi.org/10.1145/3236112.3236127>
- [72] Julian Seifert, Andreas Bayer, and Enrico Rukzio. 2013. PointerPhone: Using mobile phones for direct pointing interactions with remote displays. In *IFIP Conference on Human-Computer Interaction*. Springer, Berlin, Germany, 18–35. https://doi.org/10.1007/978-3-642-40477-1_2
- [73] Atsushi Senju and Toshikazu Hasegawa. 2005. Direct gaze captures visuospatial attention. *Visual cognition* 12, 1 (2005), 127–144. <https://doi.org/10.1080/13506280444000157>
- [74] Jun'ichiro Seyama and Ruth S Nagayama. 2007. The uncanny valley: Effect of realism on the impression of artificial human faces. *Presence: Teleoperators and virtual environments* 16, 4 (2007), 337–351. <https://doi.org/10.1162/pres.16.4.337>
- [75] Garth Shoemaker, Anthony Tang, and Kellogg S Booth. 2007. Shadow reaching: a new perspective on interaction for large displays. In *Proceedings of the 20th annual ACM symposium on User interface software and technology*. ACM, New York, NY, USA, 53–56. <https://doi.org/10.1145/1294211.1294221>
- [76] Garth Shoemaker, Takayuki Tsukitani, Yoshifumi Kitamura, and Kellogg S Booth. 2010. Body-centric interaction techniques for very large wall displays. In *Proceedings of the 6th Nordic Conference on Human-Computer Interaction: Extending Boundaries*. ACM, New York, NY, USA, 463–472. <https://doi.org/10.1145/1868914.1868967>
- [77] Mauricio Sousa, Daniel Mendes, Soraia Paulo, Nuno Matela, Joaquim Jorge, and Daniel Simões Lopes. 2017. Vrrrrroom: Virtual reality for radiologists in the reading room. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 4057–4062. <https://doi.org/10.1145/3025453.3025566>
- [78] Sophie Stellmach and Raimund Dachselt. 2012. Look & touch: gaze-supported target acquisition. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 2981–2990. <https://doi.org/10.1145/2207676.2208709>
- [79] Sophie Stellmach and Raimund Dachselt. 2013. Still Looking: Investigating Seamless Gaze-Supported Selection, Positioning, and Manipulation of Distant Targets. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Paris, France) (CHI '13)*. Association for Computing Machinery, New York, NY, USA, 285–294. <https://doi.org/10.1145/2470654.2470695>
- [80] Erik Sundén, Ingemar Lundgren, and Anders Ynnerman. 2017. Hybrid Virtual Reality Touch Table: An immersive collaborative platform for public explanatory use of cultural objects and sites. In *15th Eurographics Workshop on Graphics and Cultural Heritage, Graz, Austria, September 27-29, 2017*. Eurographics-European Association for Computer Graphics, Geneva, Switzerland, 109–113. <https://doi.org/10.2312/gch.20171300>
- [81] Hemant Bhaskar Surale, Aakar Gupta, Mark Hancock, and Daniel Vogel. 2019. TabletInVR: Exploring the Design Space for Using a Multi-Touch Tablet in Virtual Reality. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 13. <https://doi.org/10.1145/3290605.3300243>
- [82] Jayson Turner, Jason Alexander, Andreas Bulling, and Hans Gellersen. 2015. Gaze+ RST: integrating gaze and multitouch for remote rotate-scale-translate tasks. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 4179–4188. <https://doi.org/10.1145/2702123.2702355>
- [83] Boris M Velichkovsky. 1995. Communicating attention: Gaze position transfer in cooperative problem solving. *Pragmatics & Cognition* 3, 2 (1995), 199–223. <https://doi.org/10.1075/pc.3.2.02vel>
- [84] Simon Voelker, Andrii Matvienko, Johannes Schöning, and Jan Borchers. 2015. Combining direct and indirect touch input for interactive workspaces using gaze input. In *Proceedings of the 3rd ACM Symposium on Spatial User Interaction*. ACM, New York, NY, USA, 79–88. <https://doi.org/10.1145/2788940.2788949>
- [85] Daniel Vogel and Ravin Balakrishnan. 2005. Distant freehand pointing and clicking on very large, high resolution displays. In *Proceedings of the 18th annual ACM symposium on User interface software and technology*. ACM, New York, NY, USA, 33–42. <https://doi.org/10.1145/1095034.1095041>
- [86] Daniel Vogel and Ravin Balakrishnan. 2005. Distant freehand pointing and clicking on very large, high resolution displays. In *Proceedings of the 18th annual ACM symposium on User interface software and technology*. ACM, New York, NY, USA, 33–42. <https://doi.org/10.1145/1095034.1095041>
- [87] Daniel Vogel and Patrick Baudisch. 2007. Shift: a technique for operating pen-based interfaces using touch. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. ACM, New York, NY, USA, 657–666.
- [88] Julius von Willich, Markus Funk, Florian Müller, Karola Marky, Jan Riemann, and Max Mühlhäuser. 2019. You Invaded my Tracking Space! Using Augmented Virtuality for Spotting Passersby in Room-Scale Virtual Reality. In *Proceedings of the 2019 on Designing Interactive Systems Conference*. ACM, New York, NY, USA, 487–496. <https://doi.org/10.1145/3322276.3322334>
- [89] Spyros Vosinakis. 2018. Mid-Air Interaction vs Smartphone Control for First-Person Navigation on Large Displays: A Comparative Study. In *International Conference on Augmented Reality, Virtual Reality and Computer Graphics*. Springer, Berlin, Germany, 636–654. https://doi.org/10.1007/978-3-319-95282-6_45
- [90] Vanessa Vuibert, Wolfgang Stuerzlinger, and Jeremy R Cooperstock. 2015. Evaluation of docking task performance using mid-air interaction techniques. In *Proceedings of the 3rd ACM Symposium on Spatial User Interaction*. ACM, New York, NY, USA, 44–52. <https://doi.org/10.1145/2788940.2788950>
- [91] Jiayi Wang, Franziska Mueller, Florian Bernard, Suzanne Sorli, Oleksandr Sotnychenko, Neng Qian, Miguel A Otaduy, Dan Casas, and Christian Theobalt. 2020. RGB2Hands: real-time tracking of 3D hand interactions from monocular RGB video. *ACM Transactions on Graphics (TOG)* 39, 6

- (2020), 1–16.
- [92] Markus Wirth, Stefan Gradl, Jan Sembdner, Soeren Kuhrt, and Bjoern M Eskofier. 2018. Evaluation of interaction techniques for a virtual reality reading room in diagnostic radiology. In *The 31st Annual ACM Symposium on User Interface Software and Technology*. ACM, New York, NY, USA, 867–876. <https://doi.org/10.1145/3242587.3242636>
 - [93] Markus Wirth, Stefan Gradl, Jan Sembdner, Soeren Kuhrt, and Bjoern M Eskofier. 2018. Evaluation of interaction techniques for a virtual reality reading room in diagnostic radiology. In *The 31st Annual ACM Symposium on User Interface Software and Technology*. ACM, New York, NY, USA, 867–876. <https://doi.org/10.1145/3242587.3242636>
 - [94] Erroll Wood, Jonathan Taylor, John Fogarty, Andrew Fitzgibbon, and Jamie Shotton. 2016. Shadowhands: High-fidelity remote hand gesture visualization using a hand tracker. In *Proceedings of the 2016 ACM International Conference on Interactive Surfaces and Spaces*. ACM, New York, NY, USA, 77–84.
 - [95] Robert Xiao, Julia Schwarz, Nick Throm, Andrew D Wilson, and Hrvoje Benko. 2018. MRTouch: adding touch input to head-mounted mixed reality. *IEEE transactions on visualization and computer graphics* 24, 4 (2018), 1653–1660. <https://doi.org/10.1109/TVCG.2018.2794222>
 - [96] Min Xin, Ehud Sharlin, and Mario Costa Sousa. 2008. Napkin sketch: handheld mixed reality 3D sketching. In *Proceedings of the 2008 ACM symposium on Virtual reality software and technology*. ACM, New York, NY, USA, 223–226. <https://doi.org/10.1145/1450579.1450627>
 - [97] Keng-Ta Yang, Chiu-Hsuan Wang, and Liwei Chan. 2018. ShareSpace: Facilitating Shared Use of the Physical Space by both VR Head-Mounted Display and External Users. In *The 31st Annual ACM Symposium on User Interface Software and Technology*. ACM, New York, NY, USA, 499–509. <https://doi.org/10.1145/3242587.3242630>
 - [98] Difeng Yu, Hai-Ning Liang, Feiyu Lu, Vijayakumar Nanjappan, Konstantinos Papangelis, and Wei Wang. 2018. Target Selection in Head-Mounted Display Virtual Reality Environments. *J. UCS* 24, 9 (2018), 1217–1243. <https://doi.org/10.3217/jucs-024-09-1217>
 - [99] Fengyuan Zhu and Tovi Grossman. 2020. BISHARE: Exploring Bidirectional Interactions Between Smartphones and Head-Mounted Augmented Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376233>